Yanghong Liu

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EDUCATION

University of California, Berkeley — Expected Graduation: December 2026
B.A. in Statistics, Minor in Data Science — Berkeley, CA

TECHNICAL SKILLS

- **Robot Learning & Simulation:** Isaac Sim, Isaac Lab, PPO, RL, reward design
- **Robotics & Computer Vision:** Video-based 3D reconstruction, object-pose tracking, confidence estimation, teleoperation, camera calibration, policy deployment
- **LLM Training:** LoRA SFT, GRPO, ms-swift, vLLM, DeepSpeed
- **Programming & Tools:** Python, Bash, Git, Linux, R, Transformers, LaTeX

RESEARCH EXPERIENCE

UC Berkeley — Tomizuka MSC Lab (Supervised by Ruihai Wu) — Berkeley, CA
Research Intern - Robotics & Embodied AI — Dec 2025 – Present

PriDex | From Human Videos to Bimanual Dexterous Manipulation Data
UC Berkeley–General Motors Research Collaboration
Co-first author | Submitted to ICRA 2027 — Jun 2026 – Sep 2026

- **Co-developed a video-to-robot-data pipeline** transforming egocentric human videos into **physics-validated robot demonstrations** through 3D reconstruction and RL teacher rollouts with task-specific simulation success checks; recorded robot proprioception and observations from two wrist cameras and a head camera. The resulting system increased average real-world task success from **30% to 63.3% across three manipulation tasks**.
- **Led the design of confidence-guided residual PPO**, estimating position and rotation confidence via 3D-to-2D reprojection and point tracking. Used each frame's minimum score to assign high, medium, or low confidence and adapt exploration; low confidence allowed larger deviations from reconstructed references and activated a human wrist-motion direction reward to guide learning.

V2AP | Learning Affordance Posteriors for Robotic Grasping from Human Video Priors
UC Berkeley–General Motors Research Collaboration
First author | Accepted at RSS 2026 RoboData Workshop — Dec 2025 – Jun 2026

- **Led a video-to-3D reconstruction and contact-extraction pipeline** for egocentric and third-person videos, deriving human grasp priors; successful simulated grasps within human-contact regions provided **robot-executable affordance posterior labels**.
- **Led PointNet++ affordance-model training** with posterior supervision, conditioning a diffusion model on predicted affordances to generate 6-DoF grasp poses; the resulting system improved average grasp success from **31.9% to 61.6%** over human-prior supervision, evaluated on unseen object instances and unseen categories.

iDigit — Remote, China
Research Intern - Speech-to-Text & NLP — May 2023 – Aug 2023

- Built and tested a **Whisper-based speech-to-text prototype for users with hearing loss**, integrating audio input, transcription, and grammar and spelling correction into an end-to-end workflow.

COURSE PROJECT

Long-Horizon Travel Planning Agent with End-to-End Agentic RL — Mar 2026 – Jun 2026

- Built a **Qwen3-4B-Instruct ReAct travel-planning agent** integrating eight tool categories for multi-turn planning, tool use, feedback, and response generation; augmented data, generated and filtered trajectories, and applied **LoRA SFT followed by full-parameter GRPO**.
- Designed a **six-dimensional curriculum reward** for tool-use compliance, efficiency, and answer quality, using vLLM rollouts and DeepSpeed ZeRO-3 for training; on 80 test tasks, improved average LLM-judged scores by **13.46%** over the Qwen3-4B-Instruct-2507 baseline and **8.53%** over SFT.